Problem. *Cross-lingual information retrieval* (CLIR) — large amounts of information exist in low-resource languages such as Swahili or Somali. However, English monolingual users cannot easily access these resources.

Traditional approaches: translation of queries / translation of documents — depend heavily on high-quality machine translation (MT) — but for low-resource languages, MT performance is often poor.

SARAL[1] (**S**ummarization and domain-**A**daptive **R**etrieval **A**cross **L**anguages)

1. Automatic Speech Recognition — Idiap and ISI (Kaldi LF-MMI multiple models)
2. Machine Translation — Transformer + statistical syntax-based model; <2M words of parallel data + 14.5M augmented by back-translation data
   Shared Byte Pair Encoding vocabulary ≈ 8,000 tokens.
3. **Cross-Lingual Information Retrieval**
   a. Term-Level Matching: matches terms in original foreign documents and machine-translated documents; query terms are expanded by
      i. stemming,
      ii. WordNet transformations,
      iii. paraphrases,
      iv. semantic similarity.
   b. Neural Retrieval Model SEARCHER (**S**hared **E**mbedding **ARCH**itecture for **E**ffective **R**etrieval)
      Instead of translating text, it
      i. encodes queries and sentences into vectors,
      ii. computes similarity in the embedding space.

[1] What is fascinating is that *SARAL* is a Hindi word, which can be translated as *ingenious* or *simple*, depending on the relevant context.

Training data creation

Foreign Sentence S: [f1, f2, f3, f4, f5]
↓ MT
English Translations: T1, T2, T3 (multiple translations)
↓
Label:
IF any T contains query term q: S is relevant (label=1)
ELSE: S is irrelevant (label=0)

Neural architecture: **Training loop**

English Query [q1, q2, q3]	Foreign Sentence [f1, f2, f3, f4, f5]
↓	↓
Embedding Layer	CNN encoder
↓	↓
Query embeddings: matrix (E)	Contextualized Vectors F = [c1, c2, c3, c4, c5]
↓	↓

Attention mechanism
Similarity between query and sentence words: score[i,j] = E[i] · F[j]
↓
Alignment Matrix A: A[i,j] = softmax(score[i,j]) over j

0.1	0.05	0.6	0.15	0.1	← q1
0.7	0.1	0.05	0.1	0.05	← q2
0.2	0.2	0.2	0.2	0.2	← q3
f1	f2	f3	f4	f5	

↓

Query-Specific Document Representation
For each q_i — $d_i = \Sigma_j A[i,j] \times c_j$
Result: [d1, d2, d3]
Each vector captures the doc parts most relevant to each query word.
↓

Relevance Prediction: MLP
Input: [d1; d2; d3]
↓
Relevance Score p ∈ [0,1]

Loss (cross-entropy), backpropagation, update all parameters

Inference

Given:

- English Q = [q1, q2, q3]
- foreign sentences $S_1 \ldots S_n$

For each sentence:

1. Encode sentence with CNN.
2. Encode query with embedding matrix.
3. Compute attention alignment.
4. Generate vectors [$d_1 \ldots d_n$].
5. Feed into MLP → relevance score.
6. Return top N results

c. Domain Identification

   i. labeled articles (New York Times corpus),
   ii. compute domain score for n-grams: score = $count_{in_domain}$ / $count_{all}$,
   iii. a binary classifier to determine if a document belongs to the domain.

d. Summary Generation

   i. identify words or phrases related to the query inside the document
   ii. small windows of text around relevant terms

Performance

Retrieval:

AQWV **23.1** (Term-Level Matching) → **25.2** (SEARCHER)

Somali evaluation dataset: ~15 000 documents labeled (relevant / irrelevant) by native speakers of the language, 1 000 search queries

What is **AQWV**? Average Query Weighted Value:

$$AQWV = 1 - \left(\frac{Missed\ detections}{Total\ relevant\ documents} + \beta \times \frac{False\ alarms}{Total\ relevant\ documents}\right)$$

β — missing a relevant document is considered worse than showing an irrelevant one